

SIMATS ENGINEERING
(SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – SCENARIO-BASED REQUIREMENT ANALYSIS

Annexure A – Scenario-Based Requirement Analysis

Scenario 15: Smart Classroom Attendance, Engagement, and Performance Analytics System

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
Project Title:	Smart Classroom Attendance, Engagement, and Performance Analytics System		
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Date:	3-8-2026	Duration:	60 minutes

Code of Conduct

I SHAIK ARSHAD(192425080) (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ ARSHAD _____

Instructions to Students

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the assigned problem statement, perform a Scenario-Based Requirement Analysis and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. Study and Analyze the Scenario

- Carefully understand the given problem statement.
- Identify the business domain, stakeholders, existing challenges, and system objectives.

2. Identify Functional and Non-Functional Requirements

- List all functional requirements required by the proposed system.
- Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.

3. Identify Actors and Use Cases

- Identify all primary and secondary actors interacting with the system.
- List the major use cases associated with each actor.
- Draw a Use Case Diagram representing the interactions between actors and the system.

4. Prepare the Software Requirement Specification (SRS)

Develop an SRS document containing:

- Introduction
- Problem Description
- Scope of the System
- Functional Requirements
- Non-Functional Requirements
- Use Case Descriptions
- Data Requirements
- External Interface Requirements
- System Features

5. Identify Assumptions and Constraints

- List the assumptions considered while developing the system.
- Identify technical, operational, economic, legal, and time-related constraints affecting the project.

6. Validate Requirement Completeness and Consistency

- Verify whether all stakeholder requirements have been addressed.
- Check for ambiguity, redundancy, inconsistency, and missing requirements.
- Suggest improvements to enhance the quality and completeness of the requirement specification.

1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) document presents a scenario-based requirement analysis for the proposed Smart Classroom Attendance, Engagement, and Performance Analytics System. The purpose of this document is to define the functional and non-functional requirements, actors, use cases, data requirements, and constraints necessary to design and implement a reliable, scalable, and secure classroom analytics platform for educational institutions.

1.2 Industry Context

Educational institutions are increasingly adopting digital technologies to improve classroom management and student learning outcomes. Artificial Intelligence (AI), facial recognition, learning analytics, and cloud-based education platforms now enable automated attendance tracking and real-time analysis of student engagement and academic performance, replacing manual and time-consuming processes.

1.3 Intended Audience

- Software architects and developers responsible for building the system.
- Project managers and quality assurance teams.
- Academic administrators and teaching staff who will use the system.
- Evaluators and reviewers assessing this requirement analysis.

1.4 Scope of the System

The proposed system will automatically record student attendance using AI-based facial recognition or RFID technology, analyze classroom participation and engagement, continuously monitor academic performance, and generate predictive insights that help instructors identify at-risk students early. The system will provide learning analytics dashboards, support personalized learning strategies, and reduce the administrative workload associated with manual attendance and performance tracking. The scope covers the web and mobile application layers, the AI/analytics processing engine, and the cloud data storage layer, but excludes the manufacturing or procurement of physical camera/RFID hardware, which is treated as third-party equipment integrated via defined interfaces.

2. Problem Description

Traditional attendance systems and manual performance evaluation consume valuable instructional time and provide limited insight into student engagement. Roll-call attendance can take several minutes out of every class period, is prone to proxy attendance, and produces data that is rarely analyzed beyond simple presence/absence records.

Teachers often find it difficult to identify at-risk students early, since academic performance data, attendance records, and engagement indicators are usually scattered across disconnected spreadsheets, learning management systems, and paper registers. By the time a struggling student is identified through traditional methods, it is often too late for an effective intervention.

A smart classroom analytics platform should automatically record attendance using AI-based facial recognition or RFID, analyze classroom participation, monitor academic performance, and generate predictive insights for instructors. The system should help educators implement personalized interventions and improve overall learning effectiveness — forming the core problem this project addresses.

2.1 Business Domain

The system operates within the EdTech (Education Technology) domain, specifically in the sub-domain of classroom management and learning analytics for schools, colleges, and universities.

2.2 Stakeholders

Stakeholder	Role / Interest in the System
Students	Primary subjects of attendance capture and performance monitoring; consumers of personalized feedback.
Teachers / Instructors	Use engagement and performance dashboards to plan interventions and adapt teaching strategies.
Academic Administrators	Monitor institution-wide attendance and performance trends; configure policies and reports.
Parents / Guardians	Receive alerts and notifications regarding attendance and academic risk indicators.
IT / System Administrators	Maintain system infrastructure, manage user accounts, security, and integrations.
Institution Management	Use aggregated analytics for accreditation, quality assurance, and strategic decisions.

3. System Objectives and Expected Outcomes

3.1 Objectives

1. Automate student attendance recording.
2. Analyze classroom engagement.
3. Monitor academic performance continuously.
4. Predict at-risk students using AI.
5. Generate learning analytics dashboards.
6. Support personalized learning strategies.
7. Reduce administrative workload.
8. Improve teaching effectiveness.

3.2 Expected Outcomes

1. Accurate attendance management.
2. Improved student engagement.
3. Early identification of academically at-risk students.
4. Better learning outcomes.
5. Reduced manual administrative effort.
6. Data-driven teaching decisions.
7. Enhanced academic performance monitoring.
8. Increased institutional efficiency.

4. Functional Requirements

The following functional requirements define the specific behaviors and services the system must provide, derived directly from the problem statement and objectives.

ID	Functional Requirement	Priority
FR-01	The system shall capture student attendance automatically using AI-based facial recognition at the start of each class session.	High
FR-02	The system shall support RFID/smart-card based attendance as an alternative or backup identification method.	High
FR-03	The system shall flag and allow manual correction of unrecognized or ambiguous attendance entries by the instructor.	Medium
FR-04	The system shall monitor classroom engagement indicators (e.g., participation, response rate, activity completion) during a session.	High
FR-05	The system shall continuously record and update academic performance data from assessments, assignments, and quizzes.	High
FR-06	The system shall use AI/ML models to predict students at risk of falling behind academically or in attendance.	High
FR-07	The system shall generate real-time learning analytics dashboards for teachers and administrators.	High
FR-08	The system shall recommend personalized learning interventions based on individual student performance trends.	Medium
FR-09	The system shall send automated alerts/notifications to teachers, students, and parents for low attendance or declining performance.	Medium
FR-10	The system shall allow administrators to manage student, course, and class-section records.	Medium
FR-11	The system shall generate exportable attendance and performance reports (PDF/Excel) for a selected period.	Medium
FR-12	The system shall allow role-based login and access control for students, teachers, parents, and administrators.	High
FR-13	The system shall maintain a historical audit log of attendance edits and system configuration changes.	Low

5. Non-Functional Requirements

Non-functional requirements describe the quality attributes and operating constraints the system must satisfy.

Category	Requirement
Performance	The system shall recognize and record attendance for a class of 60 students within 30 seconds of session start.
Security	The system shall encrypt all biometric and personal data at rest and in transit, and comply with data protection regulations (e.g., FERPA/GDPR-equivalent).
Reliability	The system shall maintain 99.5% uptime during academic hours and gracefully fall back to manual attendance entry during outages.
Usability	The dashboard shall be usable by teachers with minimal training, requiring no more than three clicks to reach any core report.
Scalability	The system shall scale to support at least 10,000 concurrent student profiles across multiple campuses without performance degradation.
Maintainability	The system shall use a modular, service-based architecture to allow independent updates to the AI models without downtime.
Availability	Core attendance and dashboard services shall be available 24/7, excluding scheduled maintenance windows.
Accuracy	The facial recognition module shall achieve a minimum of 98% identification accuracy under normal classroom lighting conditions.
Interoperability	The system shall integrate with existing Learning Management Systems (LMS) and Student Information Systems (SIS) via REST APIs.

6. Actors and Use Cases

6.1 Actors

Actor	Type	Description
Student	Primary	Attends class, is identified by the system, and views personal attendance/performance reports.
Teacher / Instructor	Primary	Monitors engagement and performance dashboards, reviews at-risk predictions, and initiates interventions.
Administrator	Primary	Manages records, configures the system, and generates institution-wide reports.
Parent / Guardian	Secondary	Receives alerts and views summarized attendance/performance reports for their ward.
AI Engine	Secondary (external)	Performs facial recognition, engagement analysis, and at-risk prediction; supplies results to the application layer.

6.2 Use Case Summary

Use Case	Primary Actor	Description
Mark Attendance (Face/RFID)	Student, AI Engine	Captures and logs a student's presence automatically at session start.
Monitor Classroom Engagement	Teacher	Tracks participation signals during the session and summarizes engagement levels.
Track Academic Performance	Teacher, AI Engine	Aggregates assessment and assignment scores over time for each student.
Predict At-Risk Students	AI Engine	Applies a predictive model to flag students likely to underperform or disengage.
Generate Analytics Dashboard	Teacher, Administrator	Presents attendance, engagement, and performance analytics visually.
Recommend Personalized Intervention	Teacher	Suggests tailored support actions based on a student's risk profile.
Manage Student & Course Records	Administrator	Creates, updates, and maintains master data for students and courses.
Send Alerts & Notifications	System	Notifies teachers, students, or parents of attendance/performance issues.

6.3 Use Case Diagram

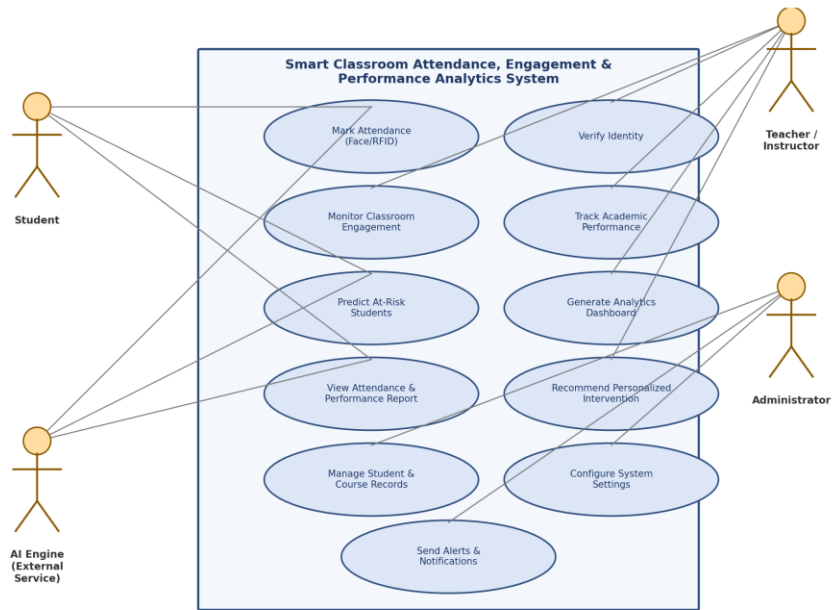


Figure 1: Use Case Diagram — Smart Classroom Attendance, Engagement & Performance Analytics System

7. System Architecture

The system follows a layered, cloud-based architecture that separates data capture, AI processing, application services, storage, and presentation, enabling each layer to scale and be maintained independently.

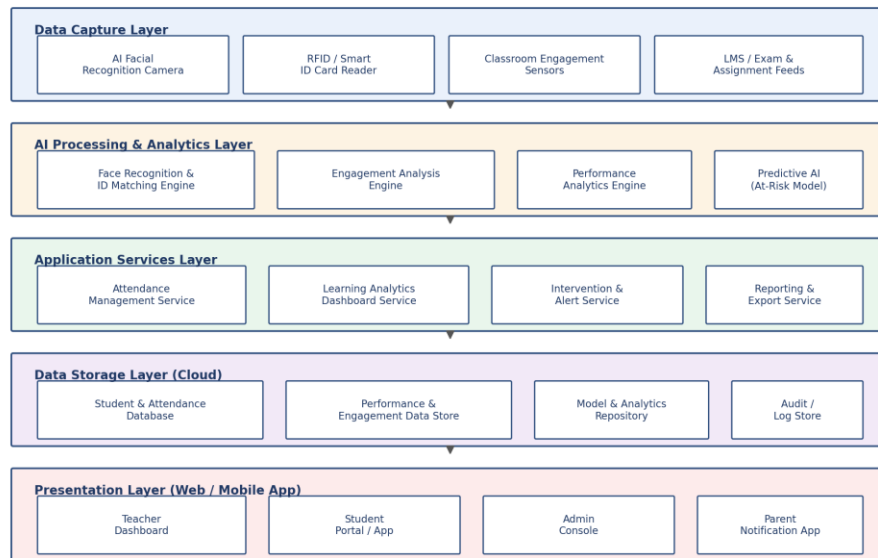


Figure 2: Layered System Architecture

7.1 Data Requirements

Data Entity	Key Attributes	Source
Student Profile	Student ID, Name, Class/Section, Face template reference, RFID tag ID	Admin console / SIS import
Attendance Record	Student ID, Session ID, Timestamp, Method (Face/RFID), Status	AI Engine, RFID reader
Engagement Log	Session ID, Student ID, Participation score, Activity completion	Classroom engagement sensors
Performance Record	Student ID, Assessment ID, Score, Date, Subject	LMS / exam feeds
Risk Prediction	Student ID, Risk score, Contributing factors, Generated date	Predictive AI model
Alert / Notification	Recipient ID, Type, Message, Timestamp, Status	Intervention & alert service

7.2 External Interface Requirements

Interface	Description
User Interface	Responsive web dashboard and mobile app for teachers, students, administrators, and parents.
Hardware Interface	Camera modules for facial recognition and RFID/NFC readers installed at classroom entry points.
Software Interface	REST APIs for integration with existing LMS (e.g., Moodle, Google Classroom) and SIS platforms.

Communication Interface	HTTPS/TLS for all client-server communication; push notification service for mobile alerts.
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8. Assumptions and Constraints

8.1 Assumptions

- Classrooms are equipped with adequate lighting and network connectivity to support AI-based facial recognition.
- Students and parents provide informed consent for biometric data collection as required by institutional policy.
- Existing LMS/SIS platforms expose usable APIs for data integration.
- Teaching staff will receive basic training to interpret analytics dashboards.
- The institution has a stable cloud/network infrastructure to host the platform.

8.2 Constraints

- Technical: Facial recognition accuracy may be affected by poor lighting, occlusion (masks), or camera angle.
- Operational: The system depends on consistent hardware maintenance (cameras/RFID readers) across classrooms.
- Economic: Institutional budget limits the initial rollout to a phased, campus-by-campus deployment.
- Legal / Compliance: Biometric data handling must comply with applicable data protection and privacy regulations.
- Time: The system must be piloted within one academic semester before full-scale institutional rollout.

9. Requirement Validation and Consistency Check

The consolidated requirement set was reviewed against the original problem statement and objectives to confirm completeness and internal consistency, and to surface gaps for improvement.

Check	Observation	Status
Stakeholder coverage	Requirements address students, teachers, administrators, parents, and IT staff.	Complete
Traceability to objectives	Each functional requirement maps to at least one stated objective (FR-01–FR-03 → attendance; FR-04 → engagement; FR-05–FR-06 → performance/prediction).	Complete
Ambiguity check	Terms such as "engagement" and "at-risk" are defined operationally via measurable indicators (participation score, risk score) to avoid ambiguity.	Addressed
Redundancy check	No duplicate requirements identified; FR-02 (RFID) is a deliberate alternative path to FR-01, not a duplicate.	Clear
Consistency check	Non-functional performance target (30 seconds/60 students) is consistent with the real-time dashboard requirement (FR-07).	Consistent
Missing requirements	Added FR-12 (role-based access) and FR-13 (audit log), which were implied by security/compliance needs but not explicit in the original scenario.	Gap closed

9.1 Suggested Improvements

- Define explicit measurable thresholds for "at-risk" classification (e.g., attendance below 75% or performance decline over two consecutive assessments).
- Add a requirement for periodic bias/fairness auditing of the facial recognition and prediction models.
- Introduce a formal data-retention and deletion policy requirement for biometric records.

10. Conclusion

This scenario-based requirement analysis translates the problem of time-consuming, low-insight classroom administration into a structured Software Requirement Specification for a Smart Classroom Attendance, Engagement, and Performance Analytics System. By combining AI-based identification, continuous performance monitoring, and predictive analytics, the proposed system directly supports the stated objectives — automating attendance, surfacing engagement trends, and enabling early, personalized intervention for at-risk students.

The functional and non-functional requirements, actor and use-case models, layered architecture, and validated data requirements presented in this document provide a sufficiently complete and consistent foundation for the subsequent design and development phases of the project.